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AUTOMATIC WASHOUT SYSTEM FOR A MIXER VEHICLE

Abstract

A concrete mixer system includes a non-transitory computer-readable medium. The non-transitory computer-readable medium stores instructions causing one or more processor to perform operations. The operations include receiving an indication that a mixing drum of a concrete mixer vehicle is about to be charged and controlling an electronically controllable valve in response to the indication such that a fluid is provided to a nozzle to facilitate pre-wetting a charge hopper of the concrete mixer vehicle prior to the mixing drum being charged.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/558,780, filed Dec. 22, 2021, which is a continuation of U.S. patent application Ser. No. 15/875,659, filed Jan. 19, 2018, which claims the benefit of U.S. Provisional Patent Application No. 62/466,960, filed Mar. 3, 2017, all of which are incorporated herein by reference in their entireties.

BACKGROUND

[0002] Concrete mixing vehicles are configured to transport wet concrete to a job site. To prevent the concrete from setting, concrete mixing vehicles include a rotatable mixing drum that continually mixes the concrete disposed therein. The drum and other components of the concrete vehicle may need to be cleaned periodically.

SUMMARY

[0003] One embodiment relates to a concrete mixer system. The concrete mixer system includes a non-transitory computer readable medium storing instructions causing one or more processors to perform operations. The operations include receiving an indication that a mixing drum of a concrete mixer vehicle is about to be charged and controlling an electronically controllable valve in response to the indication such that a fluid is provided to a nozzle to facilitate pre-wetting a charge hopper of the concrete mixer vehicle prior to the mixing drum being charged.

[0004] Another embodiment relates to a concrete mixer system. The concrete mixer system includes a non-transitory computer-readable medium storing instructions causing one or more processors to perform operations. The operations include operating a washout system in a no mess mode based on a location of a concrete mixer vehicle. The no mess mode prevents an electronically controllable valve from being activated to prevent washing at least one of a discharge hopper or a chute to prevent a fluid from being discharged onto the ground at the location.

[0005] Still another embodiment relates to a concrete mixer system. The concrete mixer system includes a non-transitory computer-readable medium storing instructions causing one or more processors to perform operations. The operations include receiving an indication that a mixing drum of a concrete mixer vehicle is about to be discharged and controlling an electronically controllable valve in response to the indication such that a fluid is provided to a nozzle to facilitate pre-wetting at least one of a discharge hopper or a chute of the concrete mixer vehicle prior to the mixing drum being discharged.

[0006] The invention is capable of other embodiments and of being carried out in various ways. Alternative exemplary embodiments relate to other features and combinations of features, as may be recited in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The invention will become more fully understood from the following detailed description taken in conjunction with the accompanying drawings wherein like reference numerals refer to like

elements, in which:

[0008] FIG. 1 is a schematic diagram of a concrete mixing truck, according to an exemplary embodiment;

[0009] FIG. 2 is a schematic diagram of a mixing drum for a concrete mixing truck, according to an exemplary embodiment;

[0010] FIG. 3 is a schematic diagram of a concrete mixing truck, according to another exemplary embodiment;

[0011] FIGS. 4 and 5 are block diagrams of an automatic washout system for a concrete mixing truck, according to an exemplary embodiment;

[0012] FIG. 6 is a detailed view of a charge hopper for a concrete mixing vehicle, including a number of nozzles mounted within the charge hopper, according to an exemplary embodiment;

[0013] FIG. 7 is another detailed view of a charge hopper for a concrete mixing vehicle, including a number of nozzles mounted within the charge hopper, according to an exemplary embodiment;

[0014] FIG. 8 is yet another detailed view of a charge hopper for a concrete mixing vehicle, including a number of nozzles mounted within the charge hopper, according to an exemplary embodiment;

[0015] FIG. 9 is yet another detailed view of a charge hopper for a concrete mixing vehicle, including a number of nozzles mounted within the charge hopper, according to an exemplary embodiment;

[0016] FIG. 10 is a detailed view of a mixing drum and a mixing element for a concrete mixing vehicle, including a number of nozzles mounted within the mixing drum, according to an exemplary embodiment;

[0017] FIG. 11 is another detailed view of a mixing drum and a mixing element for a concrete mixing vehicle, including a number of nozzles mounted within the mixing drum, according to an exemplary embodiment;

[0018] FIG. 12 is a detailed view of a mixing drum, a mixing element, a charge hopper, and a discharge hopper for a concrete mixing vehicle, including a number of nozzles mounted near the discharge hopper, according to an exemplary embodiment;

[0019] FIG. 13 is a detailed view of a mixing drum, a discharge hopper, and a chute for a concrete mixing vehicle, including a nozzles mounted underneath the discharge hopper, according to an exemplary embodiment;

[0020] FIG. 14 is another detailed view of a mixing drum, a mixing element, a charge hopper, and a discharge hopper for a concrete mixing vehicle, including a number of nozzles mounted near the discharge hopper, according to an exemplary embodiment;

[0021] FIG. 15 is yet another detailed view of a mixing drum, a mixing element, a charge hopper, and a discharge hopper for a concrete mixing vehicle, including a number of nozzles mounted near the discharge hopper, according to an exemplary embodiment;

[0022] FIG. 16 is yet another detailed view of a mixing drum, a mixing element, a charge hopper, and a discharge hopper for a concrete mixing vehicle, including a number of nozzles mounted near the discharge hopper, according to an exemplary embodiment;

[0023] FIG. 17 is a detailed view of a number of valves for a concrete mixing vehicle, according to an exemplary embodiment;

[0024] FIG. 18 is another detailed view of a number of valves for a concrete mixing vehicle, according to an exemplary embodiment;

[0025] FIG. 19 is a flow diagram of a method for controlling electronically controllable valves, according to an exemplary embodiment; and

[0026] FIGS. 20-26 are flow diagrams of a method for controlling an automatic washout system according to various modes, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0027] Before turning to the figures, which illustrate the exemplary embodiments in detail, it

should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.

[0028] According to an exemplary embodiment, a concrete mixing truck includes a mixing drum, a mixing element, a charge hopper, a discharge hopper, and a chute. The concrete mixing truck also includes a tank, an electronically controllable valve that is in fluid communication with the tank, and at least one nozzle that is fluid communication with the electronically controllable valve. The concrete mixing truck also includes an automatic washout system that controls the electronically controllable valve according to commands from a user and/or instructions from a fleet manager. The automatic washout system selectively opens at least one of the electronically controllable valves to cause the at least one nozzle to clean a portion of the concrete mixing vehicle. The nozzles cause fluid to be propelled towards a portion of the concrete mixing vehicle, thereby causing any solids on that portion to become dislodged. In some applications, fluid propelled from the nozzles is contained, or substantially contained, within the mixing drum such that no discharge of the fluid and solids, if present, to the ground occurs. The automatic washout systems can control which of the nozzles provides fluid, how much fluid is provided, and the pressure of the fluid provided, among other parameters. A number of the concrete mixing trucks may include the automatic washout system. The automatic washout systems for each of the concrete mixing trucks may be controlled together or separately. Control of the automatic washout system may be based on location of the concrete mixing truck, the type of concrete that the concrete mixing truck is transporting, the type of concrete mixing truck, and other similar parameters.

The Concrete Mixing Truck

[0029] According to the exemplary embodiment shown in FIGS. **1**, **3**, and **6-18**, a vehicle, shown as a concrete mixing truck **10**, includes a drum assembly, shown as a mixing drum **20**. According to an exemplary embodiment, the concrete mixing truck **10** is configured as a rear-discharge concrete mixing truck. In other embodiments, the concrete mixing truck **10** is configured as a front-discharge concrete mixing truck. As shown in FIG. **1**, the concrete mixing truck **10** includes a frame and a cab coupled to the frame (e.g., at a front end thereof, etc.). The mixing drum **20** is coupled to the frame and disposed behind the cab (e.g., at a rear end thereof, etc.), according to the exemplary embodiment shown in FIG. **1**. In other embodiments, such as the embodiment shown in FIG. **3**, at least a portion of the mixing drum **20** extends beyond the front of the cab. The cab may include various components to facilitate operation of the concrete mixing truck **10** by an operator (e.g., a seat, a steering wheel, hydraulic controls, a control panel, a control device, a user interface, switches, buttons, dials, etc.).

[0030] The concrete mixing truck **10** also includes an engine. For example, the engine may be coupled to the frame at a position beneath the cab. The engine may be configured to utilize one or more of a variety of fuels (e.g., gasoline, diesel, bio-diesel, ethanol, natural gas, etc.), according to various exemplary embodiments. According to an alternative embodiment, the engine additionally or alternatively includes one or more electric motors coupled to the frame (e.g., a hybrid vehicle, an electric vehicle, etc.). The electric motors may consume electrical power from an on-board storage device (e.g., batteries, ultra-capacitors, etc.), from an on-board generator (e.g., an internal combustion engine, etc.), and/or from an external power source (e.g., overhead power lines, etc.) and provide power to systems of the concrete mixing truck **10**.

[0031] The concrete mixing truck **10** may also include a transmission that is coupled to the engine. The engine produces mechanical power (e.g., due to a combustion reaction, etc.) that may flow into the transmission. The concrete mixing truck **10** may include a vehicle drive system that is coupled to the transmission. The vehicle drive system may include drive shafts, differentials, and other components coupling the transmission with a ground surface to move the concrete mixing truck **10**. The concrete mixing truck **10** may also include a plurality of tractive elements, such as wheels, that engage a ground surface to move the concrete mixing truck **10**. In one embodiment, at least a

portion of the mechanical power produced by the engine flows through the transmission and into the vehicle drive system to power at least some of the wheels (e.g., front wheels, rear wheels, etc.). In one embodiment, energy (e.g., mechanical energy, etc.) flows along a power path defined from the engine, through the transmission, and to the vehicle drive system.

[0032] As shown in FIGS. 1 and 3, the mixing drum 20 includes a mixing element (e.g., fins, etc.), shown as a mixing element 30, positioned within the interior of the mixing drum 20. The mixing element 30 may be configured to (i) mix the contents of mixture within the mixing drum 20 when the mixing drum 20 is rotated (e.g., by a drum drive system) in a first direction (e.g., counterclockwise, clockwise, etc.) and (ii) drive the mixture within the mixing drum 20 out of the mixing drum 20 (e.g., through a chute, etc.) when the mixing drum 20 is rotated (e.g., by a drum drive system) in an opposing second direction (e.g., clockwise, counterclockwise, etc.). The concrete mixing truck 10 also includes an inlet (e.g., hopper, etc.), shown as charge hopper 40, a connecting structure (e.g., a collector, a collection hopper, etc.), shown as discharge hopper 50, and an outlet, shown as chute 60. The charge hopper 40 is fluidly coupled with the mixing drum 20, which is fluidly coupled with the discharge hopper 50, which is fluidly coupled with the chute 60. In this way, wet or dry concrete may flow into the mixing drum 20 from the charge hopper 40 and wet concrete may flow out of the mixing drum 20 into the discharge hopper 50 and then into the chute 60 to be dispensed. According to an exemplary embodiment, the mixing drum 20 is configured to receive a mixture, such as a concrete mixture (e.g., cementitious material, aggregate, sand, rocks, etc.), through the charge hopper 40.

[0033] FIG. 2 illustrates the mixing drum 20, the mixing element 30, the charge hopper 40, the discharge hopper 50, and the chute 60 isolated from the concrete mixing truck 10. The mixing drum 20 may be coupled to supports (e.g., pedestals, etc.). These supports may be coupled to the frame of the concrete mixing truck 10. The supports may function to cooperatively couple (e.g., attach, secure, etc.) the mixing drum 20 to the frame and facilitate rotation of the mixing drum 20 relative to the frame. In an alternative embodiment, such as is shown in FIG. 2, the mixing drum 20 is configured as a stand-alone mixing drum that is not coupled (e.g., fixed, attached, etc.) to a vehicle. In such an embodiment, the mixing drum 20 may be mounted to a stand-alone frame. The stand-alone frame may be a chassis including wheels that assist with the positioning of the stand-alone mixing drum on a worksite. Such a stand-alone mixing drum may also be detachably coupled to and/or capable of being loaded onto a vehicle such that the stand-alone mixing drum may be transported by the vehicle.

[0034] As shown in FIG. 1, the mixing drum 20 defines a central, longitudinal axis. According to an exemplary embodiment, the mixing drum 20 is selectively rotated about this axis. The axis may be angled relative to the frame (e.g., of the concrete mixing truck 10) such that the axis intersects with the frame. For example, the axis may be elevated from the frame at an angle in the range of five degrees to twenty degrees. In other applications, the axis may be elevated by less than five degrees (e.g., four degrees, three degrees, etc.) or greater than twenty degrees (e.g., twenty-five degrees, thirty degrees, etc.). In an alternative embodiment, the concrete mixing truck 10 includes an actuator positioned to facilitate selectively adjusting the axis to a desired or target angle (e.g., manually in response to an operator input/command, automatically according to a control scheme, etc.).

The Automatic Washout System

[0035] Concrete trucks include drums that are filled with concrete at a concrete batch plant. The concrete trucks transport the concrete to a worksite and then dispense the concrete. The concrete trucks are periodically washed with fluids to prevent the buildup and formation of at least partially solidified concrete. This washing may be performed at regular intervals (e.g., daily, etc.) or after each load of concrete has been dispensed. Failure to perform regular washing could decrease the efficiency of the concrete truck or could require intensive cleaning of the concrete truck. In some applications, concrete trucks are washed with a hose (e.g., garden hose, etc.). However, this is an

arduous and lengthy process that is undesirable.

[0036] Some concrete trucks include washing systems that utilize manually-controlled valves to provide flush water for distribution (e.g., through the use of a garden hoses and sprayer, etc.). The valves receive water from a tank on the concrete truck that is typically filled at the concrete batch plant. The manually-controlled valves have to be physically opened and closed by a user (e.g., by turning a wheel, etc.), and the user manually sprays the vehicle components. It is common for operators using manually-controlled valves to use an excess amount of fluid to wash the concrete truck. Because periodic washing of the concrete truck is required, and because excess fluid is often used, the tanks on these conventional concrete trucks are often depleted quickly. As a result, the tanks may need to be refilled, potentially interrupting valuable service time provided by the concrete truck. Manual valve systems may also permit operators to deplete the water tank supply below a level needed to maintain or adjust the consistency (e.g., slump, etc.) of the concrete while en route to or at a jobsite.

[0037] Further, these conventional concrete trucks are often designed to discharge a wash mixture (e.g., waste water and concrete, etc.) onto the ground. Discharging of the fluid-concrete mixture may be regulated or prohibited in some localities. As a result, conventional concrete trucks may require intensive cleaning, may waste fluid, and may perform inefficiently and undesirably (e.g., in a costly manner, etc.) because of the use of manually-controlled valves.

[0038] As shown in FIG. 4, the concrete mixing truck **10** includes a system, shown as an automatic washout system **100**. The automatic washout system **100** facilitates cleaning of a target within the concrete mixing truck **10** (e.g., the mixing drum **20**, the mixing element **30**, the charge hopper **40**, the discharge hopper **50**, the chute **60**, etc.). The automatic washout system **100** includes a reservoir (e.g., tank, etc.), shown as source **110**, and a pump, shown as pump **120**. The source **110** is configured to store a fluid (e.g., water, non-potable water, treated water, etc.) and the pump **120** is configured to draw the fluid from the source **110**. The source **110** may be, for example, a one-hundred gallon water tank. However, the source **110** may also have other similar capacities (e.g., sixty-five gallons, ninety gallons, one-hundred and twenty-five gallons, one-hundred and fifty gallons, two-hundred gallons, two-hundred and fifty gallons, three-hundred gallons, three-hundred and eighty-five gallons, etc.). The source **110** is fluidly coupled to the pump **120**. The source **110** may be configured to selectively dose the fluid with a cleaning agent to facilitate accelerated cleaning of a target.

[0039] The automatic washout system **100** also includes a plurality of valves (e.g., solenoids, electronic valves, electronic ball valves, electromagnetic valves, etc.), shown as electronically controllable valves **130**, and a plurality of nozzles (e.g., sprayers, heads, slurry nozzles, etc.), shown as nozzles **140**. The electronically controllable valves **130** are each fluidly coupled to the pump **120**, such that fluid can be drawn from the source **110** by the pump **120** and provided to the electronically controllable valves **130**. The pump **120**, the electronically controllable valves **130**, and the nozzles **140** are connected through the use of various fluid conduits (e.g., hoses, pipes, fittings, etc.). The electronically controllable valves **130** may each selectively provide the fluid to one or more of the nozzles **140**. For example, some of the electronically controllable valves **130** may provide the fluid to two or more nozzles **140** while others of the electronically controllable valves **130** each provide the fluid to one nozzle **140**. In one embodiment, the number of electronically controllable valves **130** is equal to the number of nozzles **140**.

[0040] The nozzles **140** are each defined by a target that is provided fluid when the nozzle **140** receives fluid from the corresponding electronically controllable valve **130**. The targets of the nozzles **140** may be, for example, the mixing drum **20** (e.g., the front of the mixing drum **20**, the rear of the mixing drum **20**, etc.), the mixing element **30** (e.g., the edges of the mixing element **30**, the center of the mixing element **30**, etc.), the charge hopper **40** (e.g., an inlet of the charge hopper **40**, an outlet of the charge hopper **40**, an interior surface of the charge hopper **40**, an exterior surface of the charge hopper **40**, etc.), the discharge hopper **50** (e.g., an inlet of the discharge

hopper **50**, an outlet of the discharge hopper **50**, an interior surface of the discharge hopper **50**, an exterior surface of the discharge hopper **50**, etc.), and the chute **60** (e.g., an inlet of the chute **60**, an outlet of the chute **60**, an interior surface of the chute **60**, etc.). Each nozzle **140** may have a different target, or multiple nozzles **140** may have the same target. The nozzles **140** may be automatically or manually adjustable such that the target of each of the nozzles **140** may be tailored for a target application of the concrete mixing truck **10**.

[0041] The nozzles **140** function to remove solids (e.g., wet cement, dried cement, slurry, debris, deposits, etc.) from the targets, thereby cleaning the targets. By cleaning the targets, the automatic washout system **100** increases the longevity (e.g., service life, etc.) and desirability of the targets. In some applications, cleaning of the targets may be required by customer demands, regulatory requirements, industry standards, or other similar requirements. After the fluid is discharged by the nozzle **140** towards the target, the fluid may flow, for example, into the mixing drum **20**.

[0042] In one example, a nozzle **140** is positioned within the charge hopper **40**. When fluid is provided through this nozzle **140**, the fluid may wash solids off of the charge hopper **40**. The combination of the fluid and the solids washed from the charge hopper **40** may flow into, and be stored in, the mixing drum **20**. In another example, a nozzle **140** is positioned within the discharge hopper **50**. When fluid is provided through this nozzle **140**, the fluid may wash solids off of the discharge hopper **50**. The combination of the fluid and the solids washed from the discharge hopper **50** may be discharged onto the ground (e.g., outside of the concrete mixing truck **10**, etc.). In yet another example, a nozzle **140** is positioned within the chute **60**. When fluid is provided through this nozzle **140**, the fluid may wash solids off of the chute **60**. The combination of the fluid and the solids washed from the chute **60** may also be discharged onto the ground (e.g., outside of the concrete mixing truck **10**, etc.). In an alternative embodiment, the concrete mixing truck **10** includes a catch (e.g., gutter, reservoir, etc.) configured to collect fluid and solids discharged from the discharge hopper **50** and/or the chute **60**. In this embodiment, the catch may substantially prevent fluid and solids from being discharged onto the ground.

[0043] As shown in FIG. 5, the automatic washout system **100** includes a controller (e.g., control unit, etc.), shown as controller **150**, and an interface (e.g., human machine interface (HMI), guided user interface, software, application, etc.), shown as user interface **160**. According to some embodiments, the controller **150** is integrated within body controls (e.g., McNeilus FLEX controls, etc.) for the mixing element **30**. The controller **150** is electronically coupled to the user interface **160** and to the electronically controllable valves **130**. The user interface **160** functions to receive inputs from a user (e.g., an operator of the concrete mixing truck **10**, etc.). The controller **150** receives the inputs from the user interface **160** (e.g., via electronic communication, etc.) and provides commands to the electronically controllable valves **130**.

[0044] The controller **150** includes a circuit (e.g., module, etc.), shown as a processing circuit **152**, and a memory, shown as a memory **154**. The processing circuit **152** and the memory **154** cooperate to interpret the inputs received from the user interface **160** and to structure the commands provided to the electronically controllable valves **130**.

[0045] The controller **150** is configured to selectively engage, selectively disengage, control, and/or otherwise communicate with the electronically controllable valves **130** (e.g., actively control the components thereof, etc.). By way of example, the controller **150** may send and receive signals with the user interface **160**, the electronically controllable valves **130**, and any external device or computer. In one embodiment, the controller **150** is configured to selectively turn on and selectively turn off one or more of the various functionalities described herein. The controller **150** may turn on and turn off one or more of the various functionalities automatically, based on user requests during initial manufacture, and/or based on user input.

[0046] The controller **150** may be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a digital-signal-processor (DSP), circuits containing one or more processing components, circuitry for

supporting a microprocessor, a group of processing components, or other suitable electronic processing components. The controller **150** includes the processing circuit **152** which includes a processor, and the controller **150** also includes a memory **154**. The processing circuit **152** may include an ASIC, one or more FPGAs, a DSP, circuits containing one or more processing components, circuitry for supporting a microprocessor, a group of processing components, or other suitable electronic processing components. In some embodiments, the processing circuit **152** is configured to execute computer code stored in the memory **154** to facilitate the activities described herein. The memory **154** may be any volatile or non-volatile computer-readable storage medium capable of storing data or computer code relating to the activities described herein. According to an exemplary embodiment, the memory **154** includes computer code modules (e.g., executable code, object code, source code, script code, machine code, etc.) configured for execution by the processing circuit **152**. While the memory **154** may include various modules with particular functionality, it should be understood that the controller **150**, the processing circuit **152**, and the memory **154** may include any number of modules for completing the functions described herein. For example, the activities of multiple modules may be combined as a single module and additional modules with additional functionality may be included. Further, it should be understood that the controller **150** may further control other processes beyond the scope of the present disclosure.

[0047] The automatic washout system **100** may also include a sensor **170** that is electronically coupled to, and in electrical communication with, the controller **150**. The sensor **170** provides sensor data to the controller **150**. The controller **150** may utilize the sensor data when interpreting the inputs received from the user interface **160** and/or when structuring the commands provided to the electronically controllable valves **130**. The controller **150** may also provide the sensor data to the user interface **160**.

[0048] According to various embodiments, the user interface **160** is an electronic device (e.g., tablet, mobile electronic device, laptop, smart phone, personal electronic device, etc.). The user interface **160** may be operating an application or program that is configured to establish communication with the controller **150**. In one embodiment, the user interface **160** is a tablet running an Android application that is configured to receive inputs from a user to transmit the input (e.g., wirelessly, via near field communications, via Wi-Fi, via Bluetooth, via a network connection, via a wired connection, etc.) to the controller **150**.

[0049] In an exemplary embodiment, the user interface **160** provides the user with an image of the concrete mixing truck **10**. The image includes several selectable buttons (e.g., icons, zones, etc.) corresponding to various targets within the mixing drum **20**, within the mixing element **30**, within the charge hopper **40**, within the discharge hopper **50**, and within the chute **60**. When the user selects one of the buttons via the user interface **160**, the user interface **160** transmits an input, corresponding to the selected button, to the controller **150**. The controller **150** interprets the input to determine which targets the input corresponds with and then provides a command to the electronically controllable valve **130** associated with a nozzle **140** corresponding to the target. In this way, the user's selection on the user interface **160** facilitates a corresponding output of fluid to the target.

[0050] For each target, the user interface **160** may present the user with several options, each having a corresponding selectable button. For example, the user may be able to select various parameters such as a volume of fluid to provide to the target, an amount of time to provide fluid to the target, a pressure at which the fluid is to be provided to the target, a nozzle **140** through which the fluid is to be provided to the target, and other similar parameters. In this way, the user may utilize the user interface **160** to customize how, when, and where fluid is provided to any target or combination of targets. For example, the user interface **160** may allow the user to discharge ten gallons of water onto the mixing element **30** for thirty seconds.

[0051] In some embodiments, the user interface **160** presents the user with several selectable buttons, each corresponding to a different mode of operation of the automatic washout system **100**.

For example, the user interface **160** may present the user with a button corresponding to a “no mess” mode. Selection of this button may cause, for example, some of the electronically controllable valves **130** to open such that fluid is provided through all the nozzles **140** except the nozzles **140** corresponding to the chute **60**. Use of the “no mess” mode may facilitate cleaning of targets while minimizing or eliminating the use of excess fluid that is discharged onto the ground. The user interface **160** may also present the user with a “rapid wash” mode. In the “rapid wash” mode, the controller **150** may instruct some of the electronically controllable valves **130** to open such that fluid is provided to only the nozzles **140** required to clean the solids from the mixing element **30** and the charge hopper **40** (e.g., pre-charge, post-charge, etc.). This operation of the electronically controllable valves **130** may prevent the combination of fluid and solids from being discharged onto the ground. The “rapid wash” mode may be particularly advantageous in applications where discharge of the combination of fluid and solids is regulated or prohibited. The “rapid wash” mode may also be advantageous where it is desirable to quickly prepare the concrete mixing truck **10** to receive wet cement (e.g., from a dispenser, etc.) or for storage. The user interface **160** may provide the user with the ability to select various wash modes of operation of the automatic washout system **100**. By way of example, the automatic washout system **100** may be automatically and/or selectively operable in the “no mess” mode, the “rapid wash” mode, a “continuous wash” mode, a “wet down charge” mode, a “wet down discharge” mode, a “mix mode wash” mode, a “partial discharge wash” mode, and/or a “complete discharge wash” mode. The various washout modes of operation of the automatic washout system **100** are described with further detail herein.

[0052] In one embodiment, the user interface **160** displays prescribed limits on the volume of fluid available for the automatic washout system **100** to the user. These prescribed limits may dictate a maximum amount of fluid that the user can utilize (e.g., on a target basis, for each of the electronically controllable valves **130**, for all of the electronically controllable valves **130**, for each of the nozzles **140**, for all of the nozzles **140**, etc.). For example, the user interface **160** may display a prompt to the user stating that the mixing element **30** can only be cleaned with twenty gallons of fluid. Similar to prescribed limits on volume, the user interface **160** may display prescribed limits on the amount of time that fluid can be discharged for any or all of the electronically controllable valves **130** or nozzles **140**. In some applications, the user may decide how to allocate the fluid to comply with the prescribed limit, such as by instructing various electronically controllable valves **130** to provide fluid according to various parameters (e.g., at various pressures, for various amounts of time, to provide various amounts of fluid, in various locations, etc.). In other applications, the controller **150** may present the user with an “automatic cleaning” mode. The “automatic cleaning” mode may present the user with a recommended usage of the electronically controllable valves **130** to comply with the prescribed limit. The user may then decide to select the “automatic cleaning” mode or to change various parameters recommended by the “automatic cleaning” mode. Alternatively, the controller **150** may not allow the user to change parameters set by the “automatic cleaning” mode.

[0053] The prescribed limits (e.g., on volume, on time, etc.) may be set by a fleet manager (e.g., supervisor, etc.). The fleet manager may be actively controlled by a supervisory user or may be passively controlled by a central computer system (e.g., server). The fleet manager may manage multiple concrete mixing trucks **10**, each having the automatic washout system **100**, and may use the prescribed limits to ensure desirable and efficient operation of the concrete mixing trucks **10**. In these embodiments, the fleet manager may use a computer (e.g., computer system, personal electronic device, mobile device, smart phone, etc.) to transmit the prescribed limits to the controllers **150** in the automatic washout systems **100** for each of the concrete mixing trucks **10**.

[0054] According to an exemplary embodiment, the controller **150** provides the fleet manager with a plurality of parameters associated with the concrete mixing truck **10**. These parameters may be programmed into the memory **154**, as accessed by the processing circuit **152**, or may be obtained

by the sensor **170**, as interpreted by the processing circuit **152**. For example, the sensor **170** may determine a viscosity (e.g., a type, etc.) of concrete in the mixing drum **20**, a level of fluid in the tank, an amount of concrete in the mixing drum **20**, a location of the concrete mixing truck **10** (e.g., via a global positioning system, etc.), a fuel level of the concrete mixing truck **10**, an environmental temperature, and/or other similar parameters. The memory **154** may store, for example, configuration data associated with the concrete mixing truck **10** (e.g., fleet number, serial number, vehicle identification number, make, model, etc.), a viscosity (e.g., type, etc.) of concrete in the mixing drum **20**, maximum capacity of the mixing drum **20**, maximum capacity of the tank for fluid, maximum fuel capacity, location history (e.g., obtained from the sensor **170** and stored in the memory **154** by the processing circuit **152**), and/or other similar parameters.

[0055] Following this embodiment, the fleet manager may analyze these parameters and determine (e.g., through the use of a computer system, etc.) instructions for the controller **150** for washing the concrete mixing truck **10**. For example, the fleet manager can specify how often each target within the concrete mixing truck **10** is cleaned, in which order targets within the concrete mixing truck **10** are cleaned, how much fluid is used to clean each target within the concrete mixing truck **10**, and which of the electronically controllable valves **130** are opened and closed when each of target within the concrete mixing truck **10** is cleaned. After determining the instructions, the fleet manager can transmit the instructions to the controller **150** for a specific concrete mixing truck or for a group of concrete mixing trucks **10**. For example, the fleet manager can transmit the same instructions to all of the concrete mixing trucks **10** that are a specific model. These instructions are downloaded by the controller **150** and transmitted to the user interface **160**. In some applications, the instructions transmitted by the fleet manager can be changed on a load by load basis for the concrete mixing truck **10**. For example, if the fleet manager knows that a first load taken by the concrete mixing vehicle is to deliver a first type of concrete the fleet manager may transmit a first set of instructions whereas the fleet manager may transmit a second set of instructions when the concrete mixing vehicle is instead delivering a second type of concrete. In this way, the fleet manager can actively schedule the instructions based upon a load delivery schedule.

[0056] The user interface may provide the user with a “wash” mode, where the instructions transmitted by the fleet manager will be performed. In some instances, no other modes (e.g., the “no mess” mode, the “rapid wash” mode, etc.) are available for selection and the user can only utilize the automatic washout system **100** by selecting the “wash” mode. In some alternative applications, the “wash” mode is executed automatically by the controller **150**. In these applications, the “wash” mode may be executed based on a location of the concrete mixing truck **10**, based on an amount of fluid in the fluid tank, based on a time of day, based on an ambient temperature, or based on other similar variables, such as the parameters obtained by the sensor **170**.

[0057] FIGS. **6-20** illustrate the concrete mixing truck **10** in detail according to various embodiments. As shown in FIGS. **6-9**, the concrete mixing truck **10** includes six nozzles **140** mounted within the charge hopper **40**. The nozzles **140** are mounted against a linear interior face of the charge hopper **40**. Some of the nozzles **140** are oriented such that fluid can be propelled along the curved surface of the charge hopper **40**. Others of the nozzles **140** are directed downwards and into an aperture of the charge hopper **40** such that fluid can be propelled downwards into the mixing drum **20**. In operation, the fluid propelled from the nozzles **140** may cause solids to become dislodged from the charge hopper **40**. The combination of fluids and solids may flow through the charge hopper **40** and into the mixing drum **20**. According to one embodiment, the nozzles **140** are positioned in series along the same fluid conduit and thereby connected to the same electronically controllable valve **130**. In other embodiments, at least some of the nozzles **140** are connected to different fluid conduits and thereby may be connected to different electronically controllable valves **130**.

[0058] As shown in FIGS. **10** and **11**, the concrete mixing truck **10** includes two nozzles **140** mounted within the mixing drum **20**. The nozzles **140** extend into the mixing drum **20** and may be

oriented to direct fluid towards an inner surface of the mixing drum **20** and/or the mixing element **30**. The nozzles **140** may function to dislodge solids from the inner surface of the mixing drum **20** and/or from the mixing element **30**. In these applications, the combination of fluid and solids may be retained within the mixing drum **20**. The nozzles **140** may also function to add water to the mixing drum **20** to change the slump of concrete within the mixing drum.

[0059] Referring to FIGS. **12** and **14-16**, the concrete mixing truck **10** includes multiple nozzles **140** mounted between the charge hopper **40** and the discharge hopper **50**. These nozzles **140** may be mounted on a common mounting plate. These nozzles **140** may be oriented towards any of the mixing drum **20**, the mixing element **30**, the charge hopper **40**, the discharge hopper **50**, and the chute **60**. For example, some of the nozzles **140** may be oriented towards the discharge hopper **50** such that solids on or within the discharge hopper can be dislodged by the fluid. At least a portion of the combination of fluid and solids that flows from the discharge hopper **50** can flow through an aperture of the discharge hopper **50**, into the chute **60**, and discharged onto the ground. As shown in FIG. **16**, the nozzles **140** may each be oriented at different targets. For example, some of the nozzles **140** may be oriented towards the chute **60** such that solids on or within the chute **60** can be dislodged by the fluid. Similarly, some of the nozzles **140** may be oriented towards an outside surface of the mixing drum **20** such that solids along the outside surface of the mixing drum **20** can be dislodged by the fluid. Some of the nozzles **140** may be oriented towards an outside surface of the charge hopper **40** such that solids along the outside surface of the charge hopper **40** can be dislodged by the fluid.

[0060] As shown in FIG. **13**, the concrete mixing truck **10** includes a nozzle **140** mounted underneath the discharge hopper **50**. This nozzle **140** may be oriented towards a gap between the discharge hopper **50** and the chute **60**. This nozzle **140** may direct fluid into this gap such that solids within the gap are dislodged by the fluid. Similarly, a nozzle **140** may be oriented towards a junction between the mixing drum **20** and the discharge hopper **50**.

[0061] As shown in FIGS. **17** and **18** the electronically controllable valves **130** may be connected to a main header in parallel. In some embodiments, all of the electronically controllable valves **130** are two-way valves. However, some or all of the electronically controllable valves **130** may also be three-way valves, or any other similar valve. The electronically controllable valves **130** may include a manual override, an emergency stop, a testing function, a maintenance function, or other similar features.

[0062] According to an exemplary embodiment, the automatic washout system **100** is automatically and/or selectively operable in various washout modes including the no mess mode, the rapid wash mode, the wet down charge mode, the mix mode wash mode, the wet down discharge mode, the partial discharge wash mode, the complete discharge wash mode, and/or the continuous wash mode. The various washout modes may be selectively initiated by an operator (e.g., using the user interface **160**, manually activating the valves **130**, etc.) and/or automatically initiated by the controller **150** based on various inputs, data, washout parameters, etc.

[0063] By way of example, the no mess mode may be operator initiated to clean various targets with the nozzles **140**, while preventing water and/or concrete from exiting the mixer system and falling to the ground. For example, the electronically controllable valves **130** associated with the nozzles **140** positioned to facilitate cleaning the mixing drum **20**, the mixing element **30**, and/or the charge hopper **40** may be activated such that the respective targets are cleaned, while the nozzles **140** associated with the discharge hopper **50** and the chute **60** may remain deactivated such that water and/or concrete do not escape the mixer system through the discharge hopper **50** and the chute **60**.

[0064] By way of another example, the rapid wash mode may be operator initiated to clean various targets with the nozzles **140**. For example, the electronically controllable valves **130** associated with the nozzles **140** positioned to facilitate cleaning the mixing drum **20**, the mixing element **30**, and/or the charge hopper **40** may be activated such that the respective targets are cleaned for pre-

charging, post-charging, and/or storage.

[0065] According to an exemplary embodiment, one or more of the various washout modes of operation of the automatic washout system **100** are automatically selected, initiated, and/or terminated based on one or more washout parameters. The washout parameters may include, but are not limited to, (i) a geofence (e.g., location information such as a job site, a batch plant site, a designated washout area, etc.; using GPS and/or other location based sensing technologies; etc.), (ii) an amount of concrete discharged from the mixing drum **20** (e.g., only a portion of the concrete has been discharged, all of the concrete has been discharged, measured via sensors, estimated, etc.), (iii) the quality of the concrete within the mixing drum **20** (e.g., slump, air content, temperature, etc.; measured with sensors; estimated; etc.), (iv) environmental parameters (e.g., ambient temperature, humidity, altitude, etc.; measured with sensors; received from a weather service based on current location; etc.), (v) operator acknowledgement (e.g., the operator confirms/approves automatic initiation of a respective wash mode, etc.), (vi) mixer system component sensor readings (e.g., to determine the location, position, orientation, etc. of the components of the mixer system such as the mixing drum **20**, the chute **60**, etc.), (vii) current active functions and previously active functions (e.g., mixing, charging, discharging, driving, stopped, etc.), (viii) job type (e.g., curb, foundation, pumper, etc.), and/or (ix) information and/or commands received using telematics (e.g., wireless/remote inputs, from a fleet manager, from a server, etc.).

[0066] By way of example, the wet down charge mode may be automatically initiated by the controller **150** in response to a first subset of the washout parameters being met. Additionally or alternatively, the wet down charge mode may be manually initiated by an operator of the concrete mixing truck **10**. The controller **150** may initiate the wet down charge mode in response to (i) determining that the concrete mixing truck **10** is located at a batch plant site (e.g., determined based on GPS, a geofence, etc.), (ii) determining that the mixing drum **20** is empty (e.g., based on sensor readings, etc.), (iii) determining that the concrete mixing truck **10** is in a neutral or park gear, (iv) receiving an indication that the concrete mixing truck **10** is going to be charged with a concrete mixture (e.g., by an operator using the user interface **160**, telematically from a batch plant computer or server, etc.), and/or (v) receiving an indication that the concrete mixture will be a wet mixture, not a dry mixture (e.g., by an operator using the user interface **160**, telematically from a batch plant computer or server, etc.).

[0067] Initiating the wet down charge mode may include activating the electronically controllable valves **130** associated with the nozzles **140** positioned to facilitate cleaning and/or wetting the charge hopper **40**. The wet down charge mode may include quickly misting or lightly spraying the charge hopper **40** prior to loading or charging the mixing drum **20** with a wet mixture. Such a pre-wetting step may facilitate the wet mixture in sliding into and through the charge hopper **40** into the mixing drum **20** more easily. The amount of water sprayed into the charge hopper **40** may be controlled according to a predetermined volume threshold and/or a predetermined time threshold.

[0068] By way of another example, the mix mode wash mode may be automatically initiated by the controller **150** in response to a second subset of the washout parameters being met. Additionally or alternatively, the mix mode wash mode may be manually initiated by an operator of the concrete mixing truck **10**. The controller **150** may initiate the mix mode wash mode in response to (i) determining that the concrete mixing truck **10** is located at a batch plant (e.g., determined based on GPS, a geofence, etc.), (ii) determining that the mixing drum **20** was loaded with a concrete mixture (e.g., based on sensor readings, an operator input, etc.), (iii) determining that the concrete mixing truck **10** is in a neutral or park gear, and/or (iv) determining the mixing drum **20** is operating in a mix mode. The mixing drum **20** may be operated in the mix mode immediately after the mixing drum **20** is charged with a fresh concrete mixture or the ingredients thereof. In some embodiments, the mix mode includes rotating the mixing drum **20** at a speed of at least seven revolutions per minute (“RPM”) for up to approximately seventy revolutions.

[0069] Initiating the mix mode wash mode may include activating the electronically controllable

valves **130** associated with the nozzles **140** positioned to facilitate cleaning the charge hopper **40** and/or the interior of the mixing drum **20** (e.g., the mixing element **30** proximate the opening of the mixing drum **20**, etc.) following the charging of the mixing drum **20**. The amount of water sprayed into the charge hopper **40** and/or the interior of the mixing drum **20** may be controlled according to a predetermined volume threshold (e.g., to prevent the concrete-water ratio of the concrete mixture within the mixing drum **20** from being affected, etc.). Initiating the mix mode wash mode may additionally include activating the electronically controllable valves **130** associated with the nozzles **140** positioned to facilitate cleaning the exterior of the mixing drum **20**, the discharge hopper **50**, and/or the chute **60** following the charging of the mixing drum **20**. The amount of water sprayed into/at the discharge hopper **50**, the chute **60**, and/or the exterior of the mixing drum **20** may be controlled according to a predetermined volume threshold and/or a predetermined time threshold.

[0070] By way of yet another example, the wet down discharge mode may be automatically initiated by the controller **150** in response to a third subset of the washout parameters being met. Additionally or alternatively, the wet down discharge mode may be manually initiated by an operator of the concrete mixing truck **10**. The controller **150** may initiate the wet down discharge mode in response to (i) determining that the concrete mixing truck **10** is located at a job site (e.g., determined based on GPS, a geofence, etc.), (ii) determining that the concrete mixing truck **10** is in a neutral or park gear, (iii) determining that the chute **60** is in a discharge orientation (e.g., extended, unfolded, etc.; based on readings from a chute sensor; etc.), and/or (iv) receiving an indication that the concrete mixing truck **10** is going to discharge at least a portion of the concrete mixture (e.g., by an operator using the user interface **160**, telematically from a batch plant computer or server, telematically from a job site computer or server, etc.).

[0071] Initiating the wet down discharge mode may include activating the electronically controllable valves **130** associated with the nozzles **140** positioned to facilitate cleaning and/or wetting the discharge hopper **50** and/or the chute **60**. The wet down discharge mode may include quickly misting or lightly spraying the discharge hopper **50** and/or the chute **60** prior to discharging at least a portion of the concrete mixture from the mixing drum **20**. Such a pre-wetting step may facilitate the concrete mixture in sliding into and through the discharge hopper **50** into the chute **60** to be deposited at a desired location more easily. The amount of water sprayed into the discharge hopper **50** and/or the chute **60** may be controlled according to a predetermined volume threshold and/or a predetermined time threshold. In some embodiments, the controller **150** is configured to provide an indication to the operator in response to the chute **60** not being in the discharge orientation (e.g., based on a chute sensor output, etc.). Configuring the chute **60** in the discharge orientation may ensure proper wetting of the chute **60**.

[0072] By way of still another example, the partial discharge wash mode may be automatically initiated by the controller **150** in response to a fourth subset of the washout parameters being met (e.g., when there are one or more subsequent discharging stops for a current concrete load, pouring a couple of yards for a sidewalk and then going somewhere else to pour a few more yards for a patio pad, etc.). Additionally or alternatively, the partial discharge wash mode may be manually initiated by an operator of the concrete mixing truck **10**. The controller **150** may initiate the partial discharge wash mode in response to (i) determining that the concrete mixing truck **10** is located at, leaving, or left a job site (e.g., determined based on GPS, a geofence, etc.), (ii) determining that the mixing drum **20** discharged a portion of the concrete mixture less than a total amount of the concrete mixture within the mixing drum **20** at the job site (e.g., based on sensor readings, based on an operator input, telematically from a batch plant computer or server, telematically from a job site computer or server, etc.), and/or (iii) receiving an indication that the concrete mixing truck **10** has one or more additional discharge stops to make (e.g., based on an operator input, telematically from a batch plant computer or server, etc.).

[0073] Initiating the partial discharge wash mode may include activating the electronically

controllable valves **130** associated with the nozzles **140** positioned to facilitate cleaning the interior of the mixing drum **20** (e.g., the mixing element **30** proximate the opening of the mixing drum **20**, etc.) following the partial discharge of the mixing drum **20**. The amount of water sprayed into the interior of the mixing drum **20** may be controlled according to a predetermined volume threshold (e.g., to prevent the concrete-water ratio of the concrete mixture within the mixing drum **20** from being affected, etc.). Initiating the partial discharge wash mode may additionally or alternatively include activating the electronically controllable valves **130** associated with the nozzles **140** positioned to facilitate cleaning the exterior of the mixing drum **20**, the discharge hopper **50**, and/or the chute **60** following the partial discharge of the mixing drum **20**. The amount of water sprayed into/at the discharge hopper **50**, the chute **60**, and/or the exterior of the mixing drum **20** may be controlled according to a predetermined volume threshold and/or a predetermined time threshold. The controller **150** may be configured to control the speed at which the mixing drum **20** rotates to ensure proper cleaning of the interior and/or exterior thereof during the partial discharge wash mode. In some embodiments, the controller **150** is configured to provide an indication to the operator in response to the chute **60** not being in the discharge orientation (e.g., based on a chute sensor output, etc.). Configuring the chute **60** in the discharge orientation may ensure proper cleaning of the chute **60** (e.g., preventing concrete from getting lodged in a scam or fold of the chute **60** when stowed, folded, etc.).

[0074] By way of a further example, the complete discharge wash mode may be automatically initiated by the controller **150** in response to a fifth subset of the washout parameters being met. Additionally or alternatively, the complete discharge wash mode may be manually initiated by an operator of the concrete mixing truck **10**. The controller **150** may initiate the complete discharge wash mode in response to (i) determining that the concrete mixing truck **10** is located at, leaving, or left a job site (e.g., determined based on GPS, a geofence, etc.), (ii) determining that the concrete mixing truck **10** is on its way to or is at a batch plant site or a designated washout area (e.g., determined based on GPS, a geofence, etc.), (iii) determining that the mixing drum **20** discharged all of the concrete mixture within the mixing drum **20** (e.g., based on sensor readings, an operator input, etc.), and/or (iv) receiving an indication that the concrete mixing truck **10** has no more stops to make (e.g., based on an operator input, telematically from a batch plant computer or server, etc.).

[0075] Initiating the complete discharge wash mode may include activating the electronically controllable valves **130** associated with the nozzles **140** positioned to facilitate cleaning the interior of the mixing drum **20** (e.g., the mixing element **30** proximate the opening of the mixing drum **20**, etc.), the exterior of the mixing drum **20**, the discharge hopper **50**, and/or the chute **60** following the complete discharge of the mixing drum **20**. The amount of water sprayed into/at the discharge hopper **50**, the chute **60**, the interior of the mixing drum **20**, and/or the exterior of the mixing drum **20** may be controlled according to a predetermined volume threshold and/or a predetermined time threshold. The controller **150** may be configured to control the speed at which the mixing drum **20** rotates to ensure proper cleaning of the interior and/or exterior thereof during the complete discharge wash mode. In some embodiments, the controller **150** is configured to provide an indication to the operator in response to the chute **60** not being in the discharge orientation (e.g., based on a chute sensor output, etc.). Configuring the chute **60** in the discharge orientation may ensure proper cleaning of the chute **60** (e.g., preventing concrete from getting lodged in a scam or fold of the chute **60** when stowed, folded, etc.).

[0076] By way of still a further example, the continuous wash mode may be automatically initiated by the controller **150** in response to a sixth subset of the washout parameters being met. Additionally or alternatively, the continuous wash mode may be manually initiated by an operator of the concrete mixing truck **10**. The controller **150** may initiate the continuous wash mode in response to (i) receiving an indication (e.g., from an operator with the user interface **160**, from an external computer or server, from a sensor, etc.) that the source **110** is coupled to a continuous water source (e.g., a refilling hose, a refilling pipe, etc.), (ii) determining that the concrete mixing

truck **10** is located at a designated washout area (e.g., determined based on GPS, a geofence, etc.), (iii) determining that the concrete mixing truck **10** is in a neutral or park gear, and/or (iv) receiving an indication (e.g., from an operator with the user interface **160**, from a batch plant computer or server, etc.) of a selected region to wash (e.g., the interior of the mixing drum **20**, the exterior of the mixing drum **20**, the charge hopper **40**, the discharge hopper **50**, the chute **60**, etc.).

[0077] Initiating the continuous wash mode may include activating the electronically controllable valves **130** associated with the nozzles **140** positioned to facilitate cleaning the interior of the mixing drum **20**, the exterior of the mixing drum **20**, the charge hopper **40**, the discharge hopper **50**, and/or the chute **60**. In some embodiments, the amount of water sprayed into/at the charge hopper **40**, the discharge hopper **50**, the chute **60**, the interior of the mixing drum **20**, and/or the exterior of the mixing drum **20** is controlled according to a predetermined volume threshold and/or a predetermined time threshold. In some embodiments, the amount of water sprayed into/at the charge hopper **40**, the discharge hopper **50**, the chute **60**, the interior of the mixing drum **20**, and/or the exterior of the mixing drum **20** is not controlled based on time or volume, but rather the electronically controllable valves **130** remain open until the controller **150** receives a stop request (e.g., from the operator with the user interface **160**, from a batch plant computer or server, etc.). The controller **150** may be configured to control the speed at which the mixing drum **20** rotates to ensure proper cleaning of the interior and/or exterior thereof during the continuous wash mode. In some embodiments, the controller **150** is configured to provide an indication to the operator in response to the chute **60** not being in the discharge orientation (e.g., based on a chute sensor output, etc.). Configuring the chute **60** in the discharge orientation may ensure proper cleaning of the chute **60** (e.g., preventing concrete from getting lodged in a seam or fold of the chute **60** when stowed, folded, etc.). In some embodiments, a user operated hose is included and may be used by an operator of the concrete mixing truck **10** to provide additional cleaning to selected areas or components of the mixing system.

[0078] In some embodiments, the automatic washout system **100** is configured to activate different washout modes in succession and/or based on location and status of the concrete mixing truck **10** (e.g., the status of a transmission thereof, the status of the contents within the mixing drum **20**, etc.). By way of example, the controller **150** may detect that the mixing drum **20** has been loaded and is mixing (e.g., at an elevated speed, etc.), and the concrete mixing truck **10** is stationary (e.g., based on the transmission thereof being in a neutral gear, a park gear, etc.). Such parameters may indicate the mixing drum **20** is performing an initial mix mode to mix the contents therein and initiate a first washout mode. The first washout mode may include the mix mode wash mode. Thereafter, the controller **150** may detect that the concrete mixing truck **10** is in transit to a discharge destination (e.g., based on the transmission being in a drive gear, etc.). The controller **150** may then monitor the location of the concrete mixing truck **10** until it enters or approaches the next scheduled job site (e.g., whether the concrete mixing truck **10** enters a job site geofence, etc.). The controller **150** may then prepare for activating or activate a second washout mode in preparation for discharge. In some embodiments, the controller **150** waits to activate the second washout mode until the controller **150** receives an indication that the chute **60** is in a discharge orientation and/or the concrete mixing truck **10** is stationary (e.g., based on the transmission thereof being in a neutral gear, a park gear, etc.). The second washout mode may include the wet down discharge mode. Following discharge, the controller **150** may immediately enter a third wash mode while at the job site or wait until the concrete mixing truck **10** reaches a designated washout area (e.g., based on a washout area geofence, GPS, etc.). The third washout mode may include the partial discharge wash mode, the complete discharge wash mode, and/or the continuous wash mode.

[0079] Referring now to FIG. **19**, a method **200** for controlling the electronically controllable valves **130** is shown, according to an exemplary embodiment. At step **210**, the controller **150** receives the instructions from the fleet manager based on the parameters (e.g., location, etc.) that the controller **150** provided to the fleet manager. For example, the instructions received from the

fleet manager may specify that fluid is to be provided to the chute **60** every three hours. At step **220**, the controller **150** receives commands from the user via the user interface **160**. For example, the controller **150** may receive commands from the user interface **160** indicating that the user selected the “wash mode” previously described. At step **230**, the controller **150** opens the electronically controllable valves **130** corresponding to the instructions from the fleet manager (step **210**) and the instructions from the user (step **220**). Steps **240** and **250** occur simultaneously. At step **240**, the controller **150** continues to proceed with the fleet manager's instructions. At step **250**, the controller **150** continues to proceed with the user's commands. If the instructions or commands include prescribe limits on fluid usage (e.g., by volume, time, etc.), the controller **150** will monitor fluid usage and will close the corresponding electronically controllable valves once the prescribed limit is reached. At step **260**, the controller **150** closes at least one of the electronically controllable valve **130** in response to the prescribed limit being reached.

[0080] Referring now to FIGS. **20-26**, a method **300** for controlling a washout system (e.g., the automatic washout system **100**, etc.) according to various washout modes is shown, according an exemplary embodiment. At step **302**, a control system (e.g., the controller **150**, etc.) is configured to monitor a plurality of washout parameters during the operation of a mixer vehicle (e.g., the concrete mixing truck **10**, etc.). The plurality of washout parameters may include a geofence, an amount of concrete discharged from a mixing drum (e.g., the mixing drum **20**, etc.) of the mixer vehicle, the quality of the concrete within the mixing drum, environmental parameters, operator acknowledgement/confirmation, mixer system component sensor readings, current active functions and previously active functions, job type, information and/or commands received using telematics, and/or still other parameters.

[0081] At step **304**, the control system is configured to determine whether the plurality of washout parameters satisfy one of a first subset, a second subset, a third subset, a fourth subset, a fifth subset, or a sixth subset of the plurality of washout parameters. If none of the first subset, the second subset, the third subset, the fourth subset, the fifth subset, and the sixth subset of the plurality of washout parameters are met, the control system is configured to return to step **302**.

[0082] The control system is configured to proceed to step **310** in response to the plurality of washout parameters satisfying the first subset of the plurality of washout parameters. The first subset of washout parameters may include a geofence (e.g., the concrete mixing truck **10** is located at a batch plant site, etc.), an indication that the mixing drum is empty, an indication that the mixer vehicle is going to be charged with a concrete mixture, and/or an indication that the concrete mixture will be a wet mixture, not a dry mixture, among other possible washout parameters. At step **310**, the control system is configured to initiate a wet down charge mode of a washout system (e.g., the automatic washout system **100**, etc.). The wet down charge mode may include the control system performing steps **312** and **314**.

[0083] At step **312**, the control system is configured to activate (e.g., engage, open, etc.) valves (e.g., the electronically controllable valves **130**, etc.) associated with nozzles (e.g., the nozzles **140**, etc.) positioned to facilitate cleaning and/or wetting a charge hopper (e.g., the charge hopper **40**, etc.) of the mixer vehicle. At step **314**, the control system is configured to deactivate (e.g., disengage, close, etc.) the valves in response to the valves being open for a predetermined amount of time and/or a volume of water discharged from the nozzles exceeding a predetermined volume. Such a wet down charge mode may facilitate preparing the charge hopper to receive a wet concrete mixture such that the wet concrete mixture flows more easily therethrough into the mixing drum from the charge hopper. The control system may then be configured to return to step **302**.

[0084] The control system is configured to proceed to step **330** in response to the plurality of washout parameters satisfying the second subset of the plurality of washout parameters. The second subset of washout parameters may include a geofence (e.g., the concrete mixing truck **10** is located at a batch plant site, etc.), an indication that the mixing drum was recently loaded, an indication that the mixer vehicle is in a neutral or park gear, and/or an indication that the mixing drum is

currently operating in a mix mode, among other possible washout parameters. At step **330**, the control system is configured to initiate a mix mode wash mode of the washout system. The mix mode wash mode may include the control system performing steps **332-336**.

[0085] At step **332**, the control system is configured to activate (e.g., engage, open, etc.) valves associated with nozzles positioned to facilitate cleaning the charge hopper, a portion of an interior of the mixing drum (e.g., the mixing element **30** proximate the opening of the mixing drum **20**, etc.), an exterior of the mixing drum, a discharge hopper (e.g., a collector, the discharge hopper **50**, etc.), and/or a chute (e.g., the chute **60**, etc.) of the mixer vehicle. In some embodiments, only the valves associated with the nozzles positioned to facilitate cleaning the charge hopper, the interior of the mixing, and/or the exterior of the mixing drum are activated. At step **344**, the control system is configured to deactivate (e.g., disengage, close, etc.) the valves associated with the nozzles positioned to facilitate cleaning the charge hopper and/or the interior of the mixing drum in response to a volume of water discharged from the respective nozzles exceeding a predetermined volume (e.g., to prevent the concrete-water ratio of the concrete mixture within the mixing drum **20** from being affected, etc.). At step **346**, the control system is configured to deactivate the valves associated with the nozzles positioned to facilitate cleaning the discharge hopper, the chute, and/or the exterior of the mixing drum in response to the respective valves being open for a predetermined amount of time and/or a volume of water discharged from the respective nozzles exceeding a predetermined volume. The control system may then be configured to return to step **302**.

[0086] The control system is configured to proceed to step **350** in response to the plurality of washout parameters satisfying the third subset of the plurality of washout parameters. The third subset of washout parameters may include a geofence (e.g., the concrete mixing truck **10** is located at a job site, etc.), an indication that the chute is in a discharge orientation (e.g., extended, unfolded, etc.), and/or an indication that the mixer vehicle is going to discharge at least a portion of the concrete mixture, among other possible washout parameters. At step **350**, the control system is configured to initiate a wet down discharge mode of the washout system. The wet down discharge mode may include the control system performing steps **352** and **354**.

[0087] At step **352**, the control system is configured to activate (e.g., engage, open, etc.) valves associated with nozzles positioned to facilitate cleaning and/or wetting the discharge hopper and/or the chute of the mixer vehicle. At step **354**, the control system is configured to deactivate the valves in response to the valves being open for a predetermined amount of time and/or a volume of water discharged from the nozzles exceeding a predetermined volume. Such a wet down discharge mode may facilitate preparing the discharge hopper and/or the chute to receive a wet concrete mixture discharged from the mixing drum such that the wet concrete mixture flows more easily therethrough/therealong to a desired application area (e.g., a sidewalk, a slab, a foundation, a curb, a driveway, etc.). The control system may then be configured to return to step **302**.

[0088] The control system is configured to proceed to step **370** in response to the plurality of washout parameters satisfying the fourth subset of the plurality of washout parameters. The fourth subset of washout parameters may include a geofence (e.g., the concrete mixing truck **10** is located at, leaving, or left a job site; the concrete mixing truck **10** is at a designated washout area; etc.), an indication that the mixing drum discharged a portion of the concrete mixture less than a total amount of the concrete mixture within the mixing drum at the job site, and/or (iii) an indication that the mixer vehicle has one or more additional discharge stops to make, among other possible washout parameters. At step **370**, the control system is configured to initiate a partial discharge wash mode of the washout system. The partial discharge wash mode may include the control system performing steps **372-378**.

[0089] At step **372**, the control system is configured to activate (e.g., engage, open, etc.) valves associated with nozzles positioned to facilitate cleaning the portion of the interior of the mixing drum, the exterior of the mixing drum, the discharge hopper, and/or the chute of the mixer vehicle. At step **374**, the control system is configured to control a rotational speed of the mixing drum (e.g.,

to ensure proper cleaning of the interior and/or exterior of the mixing drum **20**, etc.). In some embodiments, the control system does not perform step **374**. At step **376**, the control system is configured to deactivate (e.g., disengage, close, etc.) the valves associated with the nozzles positioned to facilitate cleaning the interior of the mixing drum in response to a volume of water discharged from the respective nozzles exceeding a predetermined volume (e.g., to prevent the concrete-water ratio of the remaining concrete mixture within the mixing drum **20** from being affected, etc.). At step **378**, the control system is configured to deactivate the valves associated with the nozzles positioned to facilitate cleaning the discharge hopper, the chute, and/or the exterior of the mixing drum in response to the respective valves being open for a predetermined amount of time and/or a volume of water discharged from the respective nozzles exceeding a predetermined volume. The control system may then be configured to return to step **302**.

[0090] The control system is configured to proceed to step **390** in response to the plurality of washout parameters satisfying the fifth subset of the plurality of washout parameters. The fifth subset of washout parameters may include a geofence (e.g., the concrete mixing truck **10** is located at, leaving, or left a job site; the concrete mixing truck **10** is on its way back to or is at a batch plant site; the concrete mixing truck **10** is on its way to or is at a designated washout area; etc.), an indication that the mixing drum discharged all of the concrete mixture therein, and/or an indication that the concrete mixing truck **10** has no more stops to make, among other possible washout parameters. At step **390**, the control system is configured to initiate a complete discharge wash mode of the washout system. The complete discharge wash mode may include the control system performing steps **392-396**.

[0091] At step **392**, the control system is configured to activate (e.g., engage, open, etc.) valves associated with nozzles positioned to facilitate cleaning the interior of the mixing drum, the exterior of the mixing drum, the discharge hopper, and/or the chute of the mixer vehicle. At step **394**, the control system is configured to control a rotational speed of the mixing drum (e.g., to ensure proper cleaning of the interior and/or exterior of the mixing drum **20**, etc.). In some embodiments, the control system does not perform step **394**. At step **396**, the control system is configured to deactivate the valves in response to the valves being open for a predetermined amount of time and/or a volume of water discharged from the nozzles exceeding a predetermined volume. The control system may then be configured to return to step **302**.

[0092] The control system is configured to proceed to step **410** in response to the plurality of washout parameters satisfying the sixth subset of the plurality of washout parameters. The sixth subset of washout parameters may include an indication that a fluid source or tank (e.g., the source **110**, etc.) of the mixer vehicle is coupled to a continuous water source (e.g., a refilling hose, a refilling pipe, etc.), a geofence (e.g., the concrete mixing truck **10** is located at a designated washout area, etc.), an indication that the concrete mixing truck **10** is in a neutral or park gear, and/or an indication of a selected region or component of the mixer vehicle to wash with the washout system, among other possible washout parameters. At step **410**, the control system is configured to initiate a continuous wash mode of the washout system. The continuous wash mode may include the control system performing steps **412-416**.

[0093] At step **412**, the control system is configured to activate (e.g., engage, open, etc.) valves associated with nozzles positioned to facilitate cleaning the interior of the mixing drum, the exterior of the mixing drum, the charge hopper, the discharge hopper, and/or the chute of the mixer vehicle. At step **414**, the control system is configured to control a rotational speed of the mixing drum (e.g., to ensure proper cleaning of the interior and/or exterior of the mixing drum **20**, etc.). In some embodiments, the control system does not perform step **414**. At step **416**, the control system is configured to deactivate the valves in response to (i) the valves being open for a predetermined amount of time, (ii) a volume of water discharged from the nozzles exceeding a predetermined volume, and/or (iii) receiving a stop request. The control system may then be configured to return to step **302**.

[0094] The mixing drum 20 may include an injection port. The injection port may provide access into the interior of the mixing drum 20 to inject water and/or chemicals (e.g., air entrainers, water reducers, set retarders, set accelerators, superplasticizers, corrosion inhibitors, coloring, calcium chloride, minerals, and/or other concrete additives, etc.). The injection port may include a valve that facilitates injecting the water and/or the chemicals into the mixing drum 20, while preventing the mixture within the mixing drum 20 from exiting the mixing drum 20 through the injection port. In some embodiments, the mixing drum 20 includes multiple injection ports (e.g., two injection ports, three injection ports, etc.) configured to facilitate independently injecting different water and/or chemicals into the mixing drum 20.

[0095] As utilized herein, the terms “approximately,” “about,” “substantially,” and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the invention as recited in the appended claims.

[0096] It should be noted that the term “exemplary,” as used herein to describe various embodiments, is intended to indicate that such embodiments are possible examples, representations, and/or illustrations of possible embodiments (and such term is not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

[0097] The terms “coupled,” “connected,” and the like, as used herein, mean the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent) or moveable (e.g., removable, releasable, etc.). Such joining may be achieved with the two members, the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members, or the two members and any additional intermediate members being attached to one another.

[0098] References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below,” etc.) are merely used to describe the orientation of various elements in the figures. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

[0099] Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, Z, X and Y, X and Z, Y and Z, or X, Y, and Z (i.e., any combination of X, Y, and Z). Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present, unless otherwise indicated.

[0100] It is important to note that the construction and arrangement of the elements of the systems and methods as shown in the exemplary embodiments are illustrative only. Although only a few embodiments of the present disclosure have been described in detail, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements. It should be noted that the elements and/or assemblies of the components described herein may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a

wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present inventions. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the preferred and other exemplary embodiments without departing from scope of the present disclosure or from the spirit of the appended claims.

Claims

1. A concrete mixer system comprising: a non-transitory computer-readable medium storing instructions therein that, when executed by one or more processors, cause the one or more processors to: receive an indication that a mixing drum of a concrete mixer vehicle is about to be charged; and control an electronically controllable valve in response to the indication such that a fluid is provided to a nozzle to facilitate pre-wetting a charge hopper of the concrete mixer vehicle prior to the mixing drum being charged.
2. The concrete mixer system of claim 1, wherein the indication is a first indication, the electronically controllable valve is a first electronically controllable valve, and the nozzle is a first nozzle, wherein the instructions cause the one or more processors to: receive a second indication that the mixing drum is about to be discharged; and control a second electronically controllable valve in response to the second indication such that the fluid is provided to the second nozzle to facilitate pre-wetting a discharge hopper or a chute of the concrete mixer vehicle prior to the mixing drum being discharged.
3. The concrete mixer system of claim 2, further comprising the second nozzle, wherein the second nozzle is configured to be positioned proximate the discharge hopper to facilitate pre-wetting the discharge hopper prior to the mixing drum being discharged.
4. The concrete mixer system of claim 2, further comprising the second nozzle, wherein the second nozzle is configured to be positioned proximate the chute to facilitate pre-wetting the chute prior to the mixing drum being discharged.
5. The system of claim 2, wherein the instructions, when executed by the one or more processors, cause the one or more processors to operate a washout system in a no mess mode based on a location of the concrete mixer vehicle to prevent the second electronically controllable valve from being activated to prevent washing at least one of the discharge hopper or the chute to prevent the fluid from being discharged onto the ground at the location.
6. The concrete mixer system of claim 1, wherein the indication is received from at least one of an operator interface of the concrete mixer vehicle or a remote device.
7. The concrete mixer system of claim 1, further comprising the concrete mixer vehicle.
8. The concrete mixer system of claim 1, further comprising a washout system including: a tank configured to store the fluid; the electronically controllable valve; and the nozzle configured to be positioned proximate the charge hopper of the concrete mixer vehicle.
9. The concrete mixer system of claim 1, further comprising a controller including the one or more processors.
10. A concrete mixer system comprising: a non-transitory computer-readable medium storing instructions therein that, when executed by one or more processors, cause the one or more processors to operate a washout system in a no mess mode based on a location of a concrete mixer vehicle to prevent an electronically controllable valve from being activated to prevent washing at least one of a discharge hopper or a chute to prevent a fluid from being discharged onto the ground at the location.
11. The concrete mixer system of claim 10, wherein the electronically controllable valve is a first electronically controllable valve, wherein the instructions cause the one or more processors to operate the washout system in a wetdown mode that activates a second electronically controllable valve to facilitate pre-wetting a charge hopper prior to a mixing drum being charged.

- 12.** The concrete mixer system of claim 10, wherein the non-transitory computer-readable medium stores instructions therein that, when executed by the one or more processors, cause the one or more processors to operate the washout system in a wetdown mode that activates the electronically controllable valve to facilitate pre-wetting at least one of the discharge hopper or the chute prior to a mixing drum being discharged.
- 13.** The concrete mixer system of claim 10, wherein the non-transitory computer-readable medium stores instructions therein that, when executed by the one or more processors, cause the one or more processors to operate the washout system in a wash mode in response to an operator input that includes washing at least one of the discharge hopper or the chute, and wherein, when the operator input is received and the no mess mode is active, the instructions cause the one or more processors to prevent activation of the electronically controllable valve associated with the at least one of the discharge hopper or the chute, such that water and cement do not escape through the discharge hopper and the chute.
- 14.** The concrete mixer system of claim 10, further comprising the washout system including: a tank configured to store the fluid; the electronically controllable valve in fluid communication with the tank; and a nozzle fluidly coupled to the electronically controllable valve, the nozzle configured to be positioned to facilitate washing at least one of a charge hopper, a mixing drum, the discharge hopper, or the chute of the concrete mixer vehicle with the fluid.
- 15.** The concrete mixer system of claim 10, further comprising a controller including the one or more processors.
- 16.** A concrete mixer system comprising: a non-transitory computer-readable medium storing instructions therein that, when executed by one or more processors, cause the one or more processors to: receive an indication that a mixing drum of a concrete mixer vehicle is about to be discharged; and control an electronically controllable valve in response to the indication such that a fluid is provided to a nozzle to facilitate pre-wetting at least one of a discharge hopper or a chute of the concrete mixer vehicle prior to the mixing drum being discharged.
- 17.** The concrete mixer system of claim 16, wherein the indication is a first indication, the electronically controlled valve is a first electronically controlled valve, and the nozzle is a first nozzle, wherein the instructions cause the one or more processors to: receive a second indication that the mixing drum is about to be charged; and control a second electronically controllable valve in response to the second indication such that the fluid is provided to the second nozzle to facilitate pre-wetting a charge hopper of the concrete mixer vehicle prior to the mixing drum being charged.
- 18.** The concrete mixer system of claim 16, wherein the instructions, when executed by the one or more processors, cause the one or more processors to operate a washout system in a no mess mode based on a location of the concrete mixer vehicle to prevent the electronically controllable valve from being activated to prevent washing at least one of the discharge hopper or the chute to prevent the fluid from being discharged onto the ground at the location.
- 19.** The concrete mixer system of claim 16, further comprising a washout system including: a tank configured to store the fluid; the electronically controllable valve; and the nozzle configured to be positioned proximate the at least one of the discharge hopper or the chute of the concrete mixer vehicle.
- 20.** The concrete mixer system of claim 16, further comprising a controller including the one or more processors.
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